Juniper Networks® vMX Virtual Router Release Notes Release Notes

Release 19.4R1
15, July 2020

These release notes accompany the Junos OS Release 19.4R1 of the Juniper Networks® virtual MX Series router (vMX). They describe new and changed features, limitations, and known problems in the software.

Contents

Introduction

The vMX Virtual Router (vMX) is an MX Series router optimized to run on x86 servers. vMX is supported on Red Hat Enterprise Linux, CentOS, and Ubuntu.

> **NOTE:** See “System Requirements” on page 4 for details on each supported environment.

vMX enables you to leverage Junos OS Release 19.4R1 to provide a quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations as the management and configuration are the same as for MX Series routers
- Introduces new services without requiring a reconfiguration of the current infrastructure

This release of vMX supports most of the features available on Juniper Networks MX Series routers.

What’s New

There are no new features in Junos OS Release 19.4R1 for the virtual MX Series router (vMX).

- **RHEL version 7.7 support with RedHat OpenStack**—Starting with Junos OS Release 19.4R1, vMX supports Red Hat Enterprise Linux 7.7 version with RedHat OpenStack 10.

  [See Minimum Hardware and Software Requirements]

- **RHEL version 7.6 support for vMX installation on KVM**—Starting with Junos OS Release 19.4R1, you can use Red Hat Enterprise Linux 7.6 for installing vMX on KVM by running the Juniper provided installation scripts.

  [See Minimum Hardware and Software Requirements]

- **Support new instance type C5.4xlarge (vMX Instances)**—Starting in Junos OS Release 19.4R1, vMX supports EC2 instance types C5.2xlarge and C5.4xlarge on AWS. The C5.2xlarge and C5.4xlarge instance types support all the features and functionality of vMX in AWS on existing instance types. The new instance type improves network performance.

  [See Installing vMX in an AWS VPC

Known Limitations

There are no known behaviors and limitations in this release.

Open Issues

There are no open issues in this release.

Resolved Issues

This section lists the issues fixed in this release.

- After you delete the chassis license bandwidth configuration in the Agile licensing mode using the following command:

  ```
  user@host# delete chassis license bandwidth
  ```

  you must restart the vMX instance.

  PR1433157

- On vMX instances, the CPU utilization might be around 100% for a process or backup Routing Engine might crash in some specific conditions. PR1437762
- On vMX instances, incorrect interface assignment on VMware (ESXI) instances in a multi FPC chassis deployments. PR1440360
- On VMX instances and MX150, log severity level is not displayed correctly. PR1411846
- On vMX instances, IPv6 is using different unicast MAC address when you configure VRRPv3 for IPv6. PR1449014
- On VMX instances, because the host CPU core 0 is not dedicated for virtual instances, it might be used by the other applications. However, VMX's vCPU resource allocation starts from core 0. When the usage of core 0 is high, the VMX instance might get less resource, which might lead to packet drop. PR1449289
- On vMX instances, the static IP configuration of external interface in VFP VM might not be available after rebooting VFP VM. PR1451709
- On VMX platform running in SR-IOV (single-root I/O virtualization) mode with the **vlan-offload** option enabled, the **input-vlan-map** option might not work correctly. PR1460544

Licensing

Starting in Junos OS Release 19.2R1, Juniper Agile Licensing introduces a new capability that significantly improves the ease of license management network wide. The Juniper Agile License Manager is a software application that runs on your network and provides an on-premise repository of licenses that are dynamically consumed by Juniper Networks devices and applications as required. Integration with Juniper's Entitlement Management System and Portal provides an intuitive extension of the existing user experience that enables you to manage all your licenses

- The Agile License Manager is a new option that provides more efficient management of licenses, but you can continue to use individual license keys for each device if required.
- To use vMX or vBNG features licenses in Junos OS Release 19.2R1 version, you need new license keys. Previous license keys will continue to be supported for previous Junos OS releases, but for the Junos OS 19.2R1 release and later you need to carry out a one-time migration of existing licenses. Contact Customer Care to exchange previous licenses. Note that you can choose to use individual license keys for each device, or to deploy Agile License Manager for more efficient management of licenses.
- For more information about Agile Licensing keys and capabilities, see Juniper Agile Licensing portal FAQ.

See Juniper Agile Licensing Guide for more details on how to obtain, install and use the License Manager.

Upgrade Instructions

You cannot upgrade Junos OS for the vMX router from earlier releases using the **request system software add** command.

You must deploy a new vMX instance using the downloaded software package.

Remember to prepare for upgrades with new license keys and/or deploying Agile License Manager.

System Requirements

IN THIS SECTION

These topics provide system requirements for each supported environment.

Minimum Hardware and Software Requirements for KVM

Table 1 on page 5 lists the hardware requirements for KVM.

Table 1: Minimum Hardware Requirements for KVM

Description	Value
Required CPUs and NICs	For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability. For all other use cases, Intel Ivy Bridge processors or later are required. Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache To use single root I/O virtualization (SR-IOV) NIC type on vMX, the following NICs are required and no other NICs are supported with SR-IOV. • Intel X520 or X540 using 10G ports and ixgbe driver • Intel X710 or XL710 using 10G ports and i40e driver • Intel XL710-QDA2 NIC using 40G ports and i40e driver. XL710-QDA2 is only supported with i40e driver version 2.7.11 with firmware version 6.01 0x80003484 1.1747.0 or later on Ubuntu 16.04 or RHEL 7.5 When using 40G ports on the vMX instances, quality-of-service (QoS) is not supported.

Table 1: Minimum Hardware Requirements for KVM *(continued)*

Description	Value
Number of cores **NOTE:** Performance mode is the default mode and the minimum value is based on one port.	For lite mode with lab simulation use case applications: Minimum of 4 • 1 for VCP • 3 for VFP **NOTE:** If you want to use lite mode when you are running more than 3 vCPUs for the VFP, you must explicitly configure lite mode.
	For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9 • 1 for VCP • 8 for VFP The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. **NOTE:** • Without CoS—(4 * *number-of-ports*) + 4 • With CoS—(5 * *number-of-ports*) + 4 **NOTE:** All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.
Memory **NOTE:** Performance mode is the default mode.	For lite mode: Minimum of 3 GB • 1 GB for VCP • 2 GB for VFP For performance mode: • Minimum of 5 GB 1 GB for VCP 4 GB for VFP • Recommended for 16 GB 4 GB for VCP 12 GB for VFP Additional 2 GB recommended for host OS

Table 1: Minimum Hardware Requirements for KVM *(continued)*

Description	Value
Storage	Local or NAS Each vMX instance requires 30 GB of disk storage.
Other requirements	Intel VT-d capability Hyperthreading (recommended) AES-NI

Table 2 on page 7 lists the software requirements for Ubuntu.

Table 2: Software Requirements for Ubuntu

Description	Value
Operating system	Ubuntu 16.04.5 LTS (recommended host OS) Linux 4.4.0-generic
Virtualization	QEMU-KVM 2.0.0+dfsg-2ubuntu1.11
Required packages **NOTE:** Other additional packages might be required to satisfy all dependencies.	bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev libnuma-dev libyajl-dev libxml2-dev libglib2.0-dev libnl-dev python-pip python-dev libxml2-dev libxslt-dev **NOTE:** libvirt 1.3.1

Table 3 on page 7 lists the software requirements for Red Hat Enterprise Linux.

Table 3: Software Requirements for Red Hat Enterprise Linux

Description	Value
Operating system	Red Hat OpenStack Platform 10 Red Hat Enterprise Linux 7.7
Virtualization	QEMU-KVM 1.5.3

Table 3: Software Requirements for Red Hat Enterprise Linux *(continued)*

Description	Value
Required packages **NOTE:** SR-IOV requires these packages: kernel-devel gcc	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools **NOTE:** libvirt 1.2.17 or later

Table 4 on page 8 lists the software requirements for CentOS.

Table 4: Software Requirements for CentOS

Description	Value
Operating system	CentOS 7.2 Linux 3.10.0-327.22.2
Virtualization	QEMU-KVM 1.5.3
Required packages	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools **NOTE:** libvirt 1.2.19 To avoid any conflicts, install libvirt 1.2.19 instead of updating from libvirt 1.2.17.

Minimum Hardware and Software Requirements for VMware

Table 5 on page 9 lists the hardware requirements for VMware.

Table 5: Minimum Hardware Requirements for VMware

Description	Value
Number of cores **NOTE:** Performance mode is the default mode and the minimum value is based on one port.	For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9 • 1 for VCP • 8 for VFP The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. **NOTE:** To calculate the optimal number of vCPUs needed by VFP for performance mode: • Without CoS—(4 * *number-of-ports*) + 4 • With CoS—(5 * *number-of-ports*) + 4 **NOTE:** All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.
	For lite mode: Minimum of 4 • 1 for VCP • 3 for VFP **NOTE:** If you want to use lite mode when you are running with more than 3 vCPUs for the VFP, you must explicitly configure lite mode.
Memory **NOTE:** Performance mode is the default mode.	For performance mode: • Minimum of 5 GB 1 GB for VCP 4 GB for VFP • Recommended for 16 GB 4 GB for VCP 12 GB for VFP For lite mode: Minimum of 3 GB • 1 GB for VCP • 2 GB for VFP

Table 5: Minimum Hardware Requirements for VMware *(continued)*

Description	Value
Storage	Local or NAS Each vMX instance requires 30 GB of disk storage.
vNICs	• SR-IOV • VMXNET3

Table 6 on page 10 lists the software requirements.

Table 6: Software Requirements for VMware

Description	Value
Hypervisor	VMware ESXi 5.5 (Update 2), 6.0, or 6.5 **NOTE:** Due to a DPDK version change in Junos OS Release 18.1R1 or later, ESXi 6.5 is the minimum version required to run the vMX router if you are operating in high-bandwidth mode (performance mode). If you are operating the vMX router in low-bandwidth(lite) mode, you can use ESXi 6.0 or ESXi 5.5.
Management Client	vSphere 5.5 or vCenter Server

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or Partner Support Service support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf.
- Product warranties—For product warranty information, visit http://www.juniper.net/support/warranty/.
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: https://www.juniper.net/customers/support/
- Search for known bugs: https://prsearch.juniper.net/
- Find product documentation: https://www.juniper.net/documentation/
- Find solutions and answer questions using our Knowledge Base: https://kb.juniper.net/
- Download the latest versions of software and review release notes: https://www.juniper.net/customers/csc/software/
- Search technical bulletins for relevant hardware and software notifications: https://kb.juniper.net/InfoCenter/
- Join and participate in the Juniper Networks Community Forum: https://www.juniper.net/company/communities/
- Create a service request online: https://myjuniper.juniper.net

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: https://entitlementsearch.juniper.net/entitlementsearch/

Creating a Service Request with JTAC

You can create a service request with JTAC on the Web or by telephone.

- Visit https://myjuniper.juniper.net.
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see https://support.juniper.net/support/requesting-support/.

Revision History

15, July 2020—Revision 3, Junos OS Release 19.4R1—vMX

20, April 2020—Revision 2, Junos OS Release 19.4R1—vMX

26, December 2019—Revision 1, Junos OS Release 19.4R1—vMX